Global–Local Fusion With Semantic Information Guidance for Accurate Small Object Detection in UAV Aerial Images

Yaxiong Chen, *Member, IEEE*, Zhengze Ye, Haokai Sun, Tengfei Gong, Shengwu Xiong, and Xiaoqiang Lu, *Senior Member, IEEE*

***Abstract*— In recent years, the rapid development of the unmanned aerial vehicle (UAV) technology has generated a large number of aerial photography images captured by UAV. Consequently, the object detection in UAV aerial images has emerged as a recent research focus. However, due to the flexible flight heights and diverse shooting angles of UAV, two significant challenges have arisen in UAV aerial images: extreme variation in target scale and the presence of numerous small targets. To address these challenges, this article introduces a semantic information-guided fusion module specifically tailored for small targets. This module utilizes high-level semantic information to guide and align the underlying texture information, thereby enhancing the semantic representation of small targets at the feature level and subsequently improving the model's ability to detect them. In addition, this article introduces a novel global–local fusion detection strategy to strengthen the detection of small targets. We have redesigned the foreground region assembly method to address the drawbacks of previous methods that involved multiple inferences. Extensive experiments conducted on the VisDrone and UAVDT datasets demonstrate that our two self-designed modules can significantly enhance the detection capability of small targets compared with the YOLOX-M model. Our code is publicly available at: https://github.com/LearnYZZ/GLSDet.**

***Index Terms*— Decoupled head attention, feature fusion, remote sensing image recognition, robust adversarial, robustness, rotational object detection.**

Received 13 November 2024; revised 31 December 2024; accepted 14 January 2025. Date of publication 22 January 2025; date of current version 13 February 2025. This work was supported in part by Hainan Province "Nanhai New Star" Technology Innovation Talent Platform Project under Grant NHXXRCXM202361; in part by the Project of Sanya Yazhou Bay Science and Technology City under Grant NoSKJC-JYRC-2024-63, Grant SCKI-JYRC-2022-76, and Grant SKIC-2022-PTDX-031; in part by the Youth Fund Project of Hainan Natural Science Foundation under Grant 6220N344; in part by the National Natural Science Foundation of China (NSFC) under Grant 62176194 and Grant 62271484; in part by the National Key Research and Development Program of China under Grant 2022ZD0160604; in part by the Key Research and Development Program of Hubei Province under Grant 2023BAB083; in part by Hubei Natural Science Foundation for Youths under Grant 2023AFB146; in part by the Project of Sanya Science and Education Innovation Park of Wuhan University of Technology under Grant 2021KF0031 and Grant 2022KF0020; in part by MindSpore, which is a New Deep Learning Computing Framework; in part by the National Science Fund for Distinguished Young Scholars under Grant 61925112; in part by the Key Research and Development Program of Shaanxi under Grant 2023-YBGY225; and in part by the High Performance Computing Platform under Grant YZBSTCACC. *(Corresponding author: Shengwu Xiong.)*

Yaxiong Chen and Tengfei Gong are with Sanya Science and Education Innovation Park, Wuhan University of Technology, Sanya 572000, China, also with the School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan 430070, China, and also with the Interdisciplinary Artificial Intelligence Research Institute, Wuhan College, Wuhan 430212, China.

Zhengze Ye and Haokai Sun are with Sanya Science and Education Innovation Park, Wuhan University of Technology, Sanya 572000, China, and also with the School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan 430070, China.

Shengwu Xiong is with the Interdisciplinary Artificial Intelligence Research Institute, Wuhan College, Wuhan 430212, China, also with the School of Information Science and Technology, Qiongtai Normal University, Haikou 571127, China, and also with Shanghai Artificial Intelligence Laboratory, Shanghai 200232, China (e-mail: xiongsw@whut.edu.cn).

Xiaoqiang Lu is with the College of Physics and Information Engineering, Fuzhou University, Fuzhou 350108, China.

Digital Object Identifier 10.1109/TGRS.2025.3532612

I. Introduction

UNDER the strong support of policies from national and local governments for the development of the low-altitude economy, the use of aerial imaging has seen exponential growth [1]. Applying general object detection tasks to unmanned aerial vehicle (UAV) aerial images has become a new direction in the fields of computer vision and UAV technology. UAV, also known as "flying cameras," possess functions for image capture and transmission. Serving as low-altitude remote sensing platforms, they have extended into numerous practical applications, such as power line inspections, road monitoring, forest fire prevention, aerial photography, agricultural monitoring, and more. In the context of the significant development of deep learning, the object detection technology in general scene domains has made tremendous breakthroughs, achieving outstanding results on multiple benchmark datasets. Given the success in general scene domains, an increasing number of researchers are focusing on the object detection technology from the perspective of UAV based on deep learning methods [2]. However, the object detection based on UAV remote sensing platforms faces two main difficulties compared with object detection research in general scenes, due to the characteristics of aerial images captured by UAV.

1) *Numerous Small Targets*: Due to the relatively high flying altitude of UAV, targets appear in smaller sizes in the images, resulting in a higher proportion of small targets within the overall scene. This makes the identification of small targets more challenging, as shown in Fig. 1.
2) *Significant Variation in Target Scale:* The flexible flying height of the UAV results in drastic changes in the distance between the UAV and the targets. The diverse aerial shooting angles lead to images containing both small distant targets and large nearby targets simultaneously, as shown in Fig. 2.

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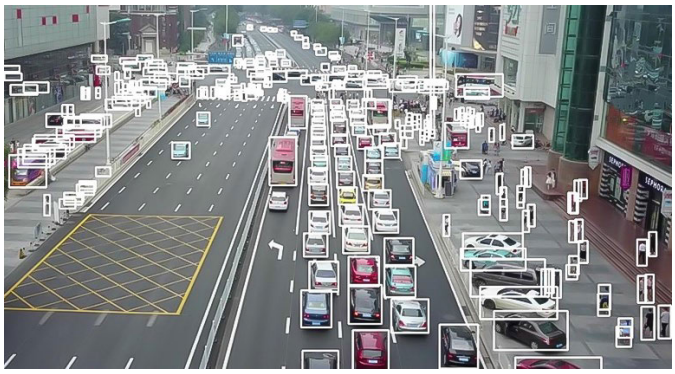


Fig. 1. This aerial photograph presents a bird's-eye view captured by a UAV, with numerous small targets scattered throughout the scene. The varying scale of these targets in the image is primarily due to the fluctuating flight altitude of the UAV during its flight.

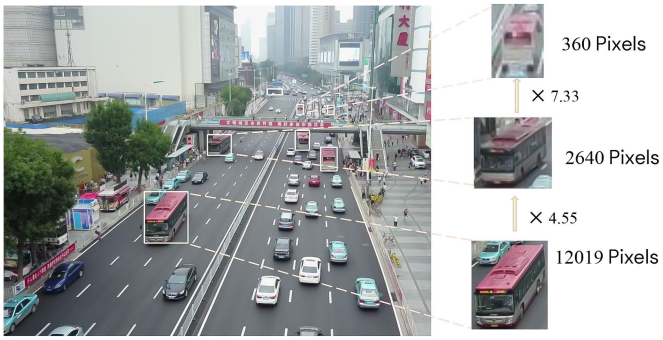


Fig. 2. You can clearly see the variation in the scale of the vehicles from the bottom right corner to the top left corner in this image.

In aerial images captured by UAV, targets often exhibit two main characteristics: significant scale variations and a higher proportion of small targets [3]. Enhancing the effectiveness of object detection from the UAV perspective and increasing detection accuracy have become a consensus among many researchers, considering these aspects. The existing research largely explores solutions from these two angles and has achieved certain results [4]. Effectively addressing the challenges of multiscale object detection and small object detection holds great value in improving the accuracy of object detection from the UAV perspective [5]. This endeavor also helps strengthen the UAV intelligent visual capabilities, laying a solid foundation for various types of applications in the future. These technical approaches can also be extended to more scenes with characteristics of multiscale and small target images, serving a broader purpose. This article will address challenges in UAV aerial imagery and aim to enhance the capability of multiscale object detection and small object detection from the perspective of UAV.

Therefore, many researchers have started by strengthening the detection of small targets and have proposed a series of solutions. Summarizing and generalizing their work, these solutions can be refined into a technical route based on global–local fusion detection [6]. It mainly consists of three parts: the global object detector responsible for global image detection, the region proposal network (RPN) or method for extracting foreground regions, and the local object detector responsible for detecting the obtained key regions [7].

Many other works have also focused primarily on how to better extract foreground regions, but their improved algorithms still face one or more of the aforementioned problems. Previous works require the introduction of additional density map generation networks [8]. Based on the image's density map, a sliding window method is used to compute the density mask, which is then further utilized to extract instance blocks [9]. A density threshold is set to exclude noisy backgrounds, and a fixed number of foreground regions are finally cropped using bounding rectangles [10]. Similarly, many hyperparameters need to be set, such as the density threshold for generating density masks and the number of foreground region blocks [11]. In addition, the quality of the density map generated by the density map generation network greatly affects the overall extraction of foreground regions. Furthermore, it involves multiple rounds of local inference.

In summary, they all share a major common flaw: significant time consumption due to multiple inferences during local detection. In addition, many methods require the introduction of additional network structures for foreground region generation. Furthermore, most of them unavoidably involve a series of hyperparameter settings, and the selection of these parameters significantly impacts the extraction results, leading to the overall detection instability. Inspired by the existing research, this article proposes a global–local fusion detection framework based on foreground region assembly. By introducing foreground region assembly technology, assembling locally detected target region blocks into a single image has changed the drawbacks of local object detectors needing to perform multiple inferences. Moreover, we have constructed a semantic information-guided module for small objects, which utilizes high-level semantic information to guide low-level texture information, thereby enhancing the feature representation of small objects and improving their detection capability. Hence, the overall solution of this section is named the global–local fusion detection framework-based small target enhancement detection algorithm.

II. Related Works

In this section, our focus is on reviewing prior research that is most relevant to the methodology we propose. This includes object detection techniques in general scenes and previous UAV object detection techniques, such as past researchers' efforts to improve small object detection and analyze the shortcomings of previous global–local fusion frameworks for foreground region assembly.

A. Object Detection in General Scenes

The remarkable advancements in deep convolutional neural networks (CNNs) have played a pivotal role in improving the performance of object detection. Most existing horizontal object detection methods can be roughly classified into two-stage and one-stage methods. The two-stage detectors, such as Faster region-based convolutional neural network (R-CNN) [12], object detection via region-based fully convolutional networks (R-FCNs) [13], and cascade R-CNN [14], first utilize a region proposal network (RPN) to generate candidate region of interests (RoIs) and then extract features from the generated regions to predict the category and spatial

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location of the RoIs. While two-stage methods yield superior detection accuracy, their adoption in industrial applications is often constrained by their computationally intensive nature and time-consuming procedures. To address the above issue, one-stage detectors, such as single shot multibox detector (SSD) [15], you only look once (YOLO) [16], and Retina-Net [17], have been proposed to directly utilize anchor boxes without the need for a separate RPN stage to predict the category and location of the objects. Although one-stage methods are more computationally efficient than two-stage counterparts, they typically tradeoff some accuracy. All the aforementioned methods are anchor-based. Recently, there has been a notable research shift toward anchor-free and transformer-based methods. For example, the fully convolutional one-stage (FCOS) object detection method in [18] tackles object detection by predicting the location of objects and regressing bounding boxes from each location. It incorporates a centerness module to effectively filter out inaccurate boxes, thereby enhancing the accuracy of object localization. The detection transformer [19] introduces the transformer for end-to-end object detection. The aforementioned methods generate horizontal bounding boxes, whereas the objects in remote sensing images are arbitrarily oriented, which results in poor detection performance. Therefore, oriented bounding boxes for accurate object detection are needed.

B. Detection Tasks for Small Objects

The small object detection is an extremely challenging task within the field of object detection, and there has been a considerable amount of research dedicated to it. Summarizing these efforts, the main difficulties of small object detection can be categorized as follows.

1) The feature representation tends to be insufficient. Because small objects cover a limited number of pixels, this implies that the features available for classification and localization are limited.
2) In deeper feature extraction networks, they are susceptible to the influence of feature noise, leading to even more inadequate feature representation. Moreover, they are highly sensitive to position offset, where slight disturbances can result in a significant decrease in intersection over union (IoU), which is currently the basis for evaluation metrics in object detection, thus leading to suboptimal detection results.

Existing research explores methods such as super-resolution, feature fusion, and information enhancement from the perspective of enhancing feature representation [20]. People address the issue by employing data augmentation to increase the total quantity and richness of small object samples [13], [21] or by magnifying the detection of local object regions [22], [23]. The following sections will discuss and analyze the current research status around these two main aspects.

1) Enhanced Feature Representation: Increasing the resolution of features can compensate for the loss of information in small objects. Therefore, super-resolution methods have been introduced into object detection tasks to enhance the resolution of small objects, achieving promising results. Li et al. [24] proposed a perceptual generative adversarial network that explores the intrinsic structural relationships between objects of different sizes, thus enhancing the feature representation of small objects. They applied this model to traffic sign detection and pedestrian detection. Bai et al. [25] introduced a multitask generative adversarial network where the generator consists of a super-resolution network responsible for upsampling, generating higher resolution images from smaller blurry images. The discriminator is a multitask network that refines the classification and localization results of detection boxes based on the original task of discriminating image authenticity. Noh et al. [26] found that matching the receptive fields of high-resolution features with the original input and target features could enhance the supervision of super-resolution networks. Accordingly, they introduced dilated convolutions into the instance feature extractor to match the receptive fields and directly supervised the super-resolution generator. However, these super-resolution methods are limited to relying solely on low-resolution feature maps to imagine and supplement missing details, overlooking important contextual information contained in different scale features extracted by backbone networks. Therefore, these methods tend to generate false textures and details, leading to false positives in feature generation. In addition, super-resolution-based object detection methods often have more complex network structures, increased training difficulty, and more uncontrollable factors.

Addressing the issue of insufficient feature representation for small objects, besides super-resolution methods, many works have also been explored from other perspectives. Zhang et al. [27] constructed a channelwise perceptual deconvolution module to reorganize and fuse feature maps from different channels to obtain richer semantic information, based on which a content-aware object detector was designed. Liu et al. [28] employed a feature-enhancing operation strategy to gradually incorporate information from higher layers to enhance feature representation, and experimental results showed that a small number of candidate region boxes could achieve a high recall rate for the RPN. Hu et al. [29] fully utilized feature information from different scales to design a detector specifically for small objects, using multiscale feature fusion to ensure that small objects have sufficient detail and semantic features. Cao et al. [30] adopted a similar strategy based on SSD, introducing contextual information and achieving feature fusion through feature concatenation and addition operations, surpassing the accuracy of deconvolutional methods while ensuring certain inference speed. Guan et al. [31] achieved precise candidate region extraction by constructing feature maps with higher resolution and fusing contextual information from multiscale features through pyramid pooling operations to obtain stronger context-aware features. Concurrently, Fang and Shi [32] also introduced a flexible context information fusion module from the perspective of enhancing contextual information. Liu et al. [33] proposed the high-resolution detection network (HRDet), leveraging the advantages of high-resolution features to improve the detection performance of small objects, consisting of multiple deep pyramid networks and multiscale feature pyramid networks (FPNs). Among them, the multi-depth pyramid network contains a group of backbone networks with different depths, sending high-resolution feature maps to shallower backbone networks to preserve more location

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information while inputting low-resolution feature maps into deeper backbone networks to extract more semantic information. Qiang Li's enhanced global pyramid network (EGPNet) enhances the detection of infrared small targets through multiscale feature fusion, edge-guided image refinement, and local target amplification techniques. Experimental results show that the model effectively detects small and weak targets. Qiang Li's EGPNet [34] enhances the detection of infrared small targets through multiscale feature fusion, edge-guided image refinement, and local target amplification techniques. Experimental results show that the model effectively detects small and weak targets. Finally, various features from different scales are fused to obtain high-resolution feature maps with more critical information. In summary, multiscale feature fusion helps capture richer semantic and detailed information to aid better classification and localization. However, if the fusion process is poorly controlled, the generation of redundant information may introduce noise to the feature maps of small objects, causing distortion and leading to the performance degradation. Therefore, the effective information aggregation is crucial.

2) Data Augmentation Strategies for Small Objects: Feature fusion is the process of combining features from different sources or representations. Its effectiveness lies in its ability to comprehensively utilize information from different features, thereby improving model performance and generalization capability. Feature fusion can help the model capture richer data representations, thus enhancing task accuracy and robustness. The feature fusion concatenates multiple feature maps along the depth dimension, making the feature representation more comprehensive. For example, in encoders and decoders, concatenating low-level encoder features with high-level decoder features enhances the decoder's perceptual capability. The pyramid pooling involves pooling features of different scales to obtain multiscale information, thereby improving model performance. In addition, the deconvolutional operation upsamples high-level features to the same resolution as low-level features and then concatenates the two feature maps to obtain more accurate results. For example, FPN is used for object detection and semantic segmentation tasks, merging feature information from different levels to generate high-resolution, high-semantic feature maps. Atrous spatial pyramid pooling (ASPP): utilizes the feature pyramid fusion method with dilated convolutions to enhance the perception of different scales and semantic information. Dual attention network (DANet): combines channel and spatial attention mechanisms to enhance semantic segmentation performance. Graph convolutional network (GCNet): utilizes global context information to enhance the expression ability of local feature maps. However, feature fusion methods do not increase the expression ability of small targets or the proportion of related feature layers.

Furthermore, research on data augmentation strategies, as an important branch in small object detection, has also yielded promising results [13], [21], [23]. Kisantal et al. [13] analyzed the poor performance of small object detection in general scene domains from the perspective of sample data. They observed that in general scene datasets: the number of images containing small objects is small, and among these images, both the overall quantity and coverage area of small objects are limited, causing models to focus more on learning medium to large-sized objects. Therefore, they proposed a solution to oversample and enhance images containing small objects. Specifically, for images containing small objects, they randomly selected small objects and copied them multiple times at arbitrary positions in the image to increase the number of small object samples. They conducted extensive experiments to determine the optimal values for oversampling and copy–paste operations, effectively improving the detection performance of small objects but at the cost of decreased performance for medium to large-sized objects. In addition, to enhance the detection of small objects, Akyon et al. [23] proposed an open-source general slicing-assisted inference framework that can be easily integrated into existing object detectors. They divided the original image into multiple overlapping image blocks, scaled them according to their aspect ratios, and fed them separately into the object detection model for prediction. Finally, they aggregated all prediction results and those of the original image and obtained the final output through a nonmaximum suppression (NMS) operation. Akyon et al. [23] applied the slicing inference technique to object detection in aerial images captured by UAV to increase the accuracy of small object detection. Although this method significantly improves the detection performance of small objects, it incurs substantial inference overhead and is sensitive to the settings of hyperparameters, such as image slicing size and overlap ratio.

C. Global–Local Fusion Detection Framework

Enhancing the detection effectiveness of small targets in aerial images captured by UAV addresses one of the major challenges in improving object detection performance from the perspective of UAV. Researchers have proposed a series of solutions following a "global–local" fusion detection process. The overall framework consists of three main parts: an RPN or method for extracting foreground regions, a global object detector, and a local object detector. Foreground region blocks are obtained from aerial images using RPNs or related algorithms; then, they are individually processed by the local object detector for detection. The results are fused with those obtained from the entire image by the global object detector to obtain the final prediction.

Yang et al. [35] proposed clustered detection network (ClusDet) as a highly representative work in this research direction, which is very inspiring. It mainly includes a cluster proposal subnetwork (CPNet) [36] for extracting clustered regions (foreground regions), a ScaleNet for scale estimation, and two object detectors, respectively, targeting local regions and the entire image. The clustered regions are foreground regions where targets are densely clustered in the image. The design of the cluster proposal network draws inspiration from the construction of candidate region networks in multistage object detectors. However, because large regions are predicted, a larger receptive field is required, thus deep features extracted from backbone networks are selected as input. Moreover, as a fully supervised method, manual strategies are required to generate corresponding training labels for clustered regions. To address the issue of dense and overlapping results produced by CPNet, a cluster region iteration algorithm is designed,

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which further optimizes the extracted cluster regions using a method similar to NMS. Then, a scale estimation network is introduced to estimate the scale of cluster regions, and methods, such as padding and segmentation, are used to adjust the cluster regions to avoid extreme-scale cluster regions. Finally, the local detection results of clustered regions and the detection results of the entire image are processed through NMS to obtain the final fused prediction. ClusDet significantly enhances the detection effectiveness of small targets. However, it also has several drawbacks: its network structure is relatively complex, increasing training time and difficulty. The true labels for CPNet need to be manually generated, and the number of cluster regions is fixed and manually set, which does not reflect the flexible and changeable nature of target aggregation areas in the UAV perspective. In addition, processing clustered regions separately for detection increases overall inference time.

III. Proposed Method

This article proposes a semantic information-guided fusion module specifically for small targets. The module utilizes high-level semantic information to guide and align the underlying texture information, thereby enhancing the semantic representation of small targets at the feature level. We are able to generate high-resolution, information-rich feature maps, effectively improving the feature representation and localization accuracy of small targets. In addition, this article introduces a novel global–local fusion detection strategy to strengthen the detection of small targets. We have redesigned the foreground region assembly method to address the drawbacks of previous approaches involving multiple inferences, making the global and local detectors more adaptive.

A. Semantic-Guided Feature Fusion for Small Targets

The small object detection is mainly challenging due to its insufficient feature representation, stemming primarily from two aspects: First, small targets occupy fewer pixels, resulting in smaller regions in the feature maps related to them, thus limiting the availability of effective features for classification and localization. Second, in deep-level feature extraction networks, noise disturbances may occur, further reducing the adequacy of feature representation. Therefore, a small target texture information capture extension layer is designed, selecting low-level, feature-sparse high-resolution images as the foundation. Drawing inspiration from super-resolution techniques and considering the contextual relationships between features of different scales, a feature fusion enhancement module is devised. This module aims to obtain higher resolution, more information-rich feature maps, enabling better feature representation in the extension detection layer for small targets. In addition, the receptive field of the extension detection layer is smaller compared with other layers, aiding in more accurate determination of the small target's position, thereby facilitating better classification and localization by the corresponding detection head.

1) Semantic Information Extraction for Small Objects: After adding the small target texture information capture extension layer, the overall network changes from three detection heads to four detection heads. From bottom to top, the input feature maps for each detection head are denoted by P_1–P_4, with corresponding sizes of $(H/4) \times (W/4)$, $(H/8) \times (W/8)$, $(H/16) \times (W/16)$, and $(H/32) \times (W/32)$, where H and W represent the height and width of the input image, respectively. Among them, $\{P_2, P_3, P_4\}$ are obtained by the path aggregation feature pyramid network (PAFPN) based on the deeper three-layer features extracted by the backbone network through two fusion paths: top–down and bottom–up, with channel numbers of $2C$, $4C$, and $8C$, respectively. P_1 is based on the shallow and high-resolution feature map C_1 of the backbone network, with P_2 and P_3 as supplementary sources of key information, obtained through the designed feature fusion enhancement module. Compared with the original feature map C_1, P_1 has the same large size but contains richer semantic and local detail information. Similar to Section II, since the feature map C_1 is the shallow feature extracted by the backbone network, its feature information are relatively rough. Therefore, it is passed through a cross-stage partial (CSP) module for feature refinement and channel expansion, generating C_1' with a channel number of $2C$. The generation process of P_1 can be mathematically expressed as follows:

$$P_1 = C_1' + Up_{2\times}(P_2'). \tag{1}$$

Here, $Up_{2\times}(\cdot)$ denotes $2\times$ upsampling using nearest neighbor interpolation. P_2' is obtained through fusion operations from the upper two layers P_2 and P_3, and then upsampled to ensure the same size as C_1'.The generation process of P_2' is the main part of the feature fusion enhancement module. The core of this module is to effectively extract semantic information from low-resolution feature maps and important local details from high-resolution maps, enabling the efficient capture of key features and thus assisting in generating high-resolution feature maps with better feature representation. The collective process is illustrated in Fig. 3.

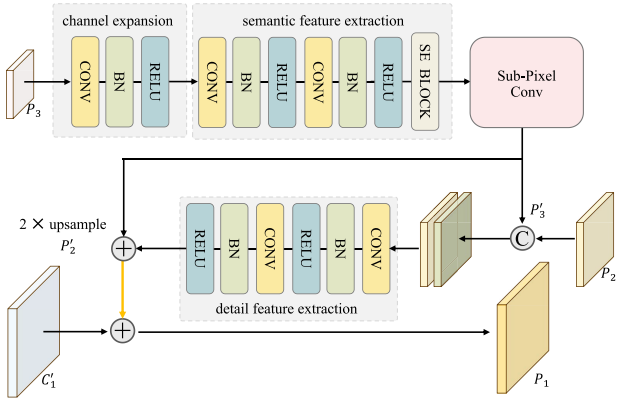


Fig. 3. P_1 utilizes the shallow, high-resolution feature map C_1 from the backbone network as the foundation. In addition, P_2 and P_3 generated by PAFPN serve as supplementary sources for critical information. These are obtained through the designed feature fusion enhancement module. The specific process involves P_3 undergoing channel dilation, semantic feature extraction, and subpixel cleansing operations to obtain P_1'. P_1' is then concatenated with and undergoes further detail feature extraction to yield P_2'. Finally, P_2' is fused with C_1 through double upsampling to produce P_1.

2) High-Level Feature Semantics Guiding Low-Level Features: The design of the semantic feature extraction module consists of two consecutive convolutional blocks and one squeeze-and-excitation (SE) attention residual block [37].

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In this section, we introduce a channel attention mechanism, which can adaptively learn the importance of each channel, to prepare for the next upsampling operation, facilitating the generation of better high-resolution feature maps. The SE attention process mainly includes two stages: first, using global average pooling to compress the feature map into a 1-D vector, and then using a set of fully connected layers to explore the important relationships of different channels, using the Sigmoid activation function to generate weights for different channels, and finally, weighting the original feature map channelwise to enhance the feature expression in the channel dimension. For the spatial scale increase of P_3', this section uses subpixel convolution to complete the upsampling. Subpixel convolution was proposed by Shi et al. [38] and has been used in image super-resolution reconstruction tasks with excellent results. Compared with other upsampling methods (deconvolution and interpolation), subpixel convolution [38] has two main advantages: it can reduce computational complexity and memory consumption. It can fully utilize high-dimensional channel information to achieve more accurate and refined spatial feature representation. Pixel shuffle operation is used in subpixel convolution to increase the size of the feature map, as shown in Fig. 4, transforming the feature map F of size $H \times W \times u^2C$ into a feature map F' of size $uH \times uW \times C$ through pixel reorganization. The mathematical formula (3) below can describe this operation process

$$\mathcal{PS}(F)_{x,y,c} = F_{\lfloor \frac{x}{u} \rfloor, \lfloor \frac{y}{u} \rfloor, C*u*\mathrm{mod}(y,u)+C\times mo(x,u)+c}. \tag{2}$$

$\mathcal{PS}(F)_{x,y,c}$ represents the value of the pixel at position (x, y, c) on the new feature map F' after the pixel shuffle operation $\mathcal{PS}(\cdot)$. u corresponds to the upsampling factor, set to 2 in this section. $\lfloor \bullet \rfloor$ denotes the floor function, and $\mathrm{mod}(a, b)$ indicates the remainder of a divided by b. In this article, P_3' is concatenated with P_2' along the channel dimension to ensure the richness of detail information. Then, they are fed into the detail feature extraction module to learn local key texture features and detail features. Similar to the semantic feature extraction module, the detail feature extraction module consists of two consecutive convolutional blocks. Ultimately, the generation of the larger resolution feature map P_2' with stronger semantic features and local detail features can be expressed as follows:

$$P_2' = P_3' + \mathrm{EF}_d\left(\mathrm{Cat}\left(P_2, P_3'\right)\right). \tag{3}$$

$\mathrm{EF}_d(\cdot)$ corresponds to a detail feature extraction module and Cat $(\cdot)$ corresponds to feature concatenation operation. Through the feature fusion enhancement module, high-resolution feature maps containing rich semantic and detail information can be obtained in a relatively short time as input to the expanded detection layer. This allows small targets to have richer information expression, thereby alleviating the problem of decreased detection performance due to the suboptimal feature representation [39]. This helps to more accurately localize and predict the categories of small targets.

B. Global–Local Fusion Detection Framework Based on Foreground Region Assembly

By introducing an expanded detection layer and utilizing the feature fusion enhancement module, high-resolution and information-rich feature maps are generated to achieve a better representation of relevant features for small targets. Building upon the object detection method proposed in the preceding section, this section takes a different approach by adopting a global–local fusion detection strategy to enhance the detection of small targets [40]. For local detection, foreground regions are extracted and then scaled to amplify the clustered areas of small targets, thereby reducing the difficulty of detecting small targets [41]. Compared with directly enlarging the input image size, this local detection strategy can more precisely and specifically amplify key regions, and by scaling down the proportion of small targets, it improves the effectiveness of local object detection. In addition, it effectively reduces the time required for inference.

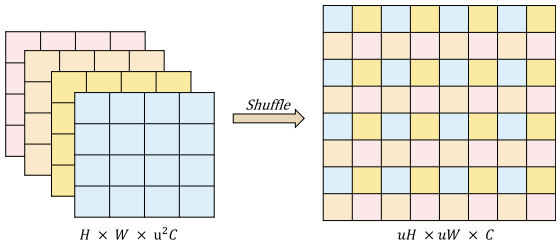


Fig. 4. This is the subpixel cleansing operation. Subpixel cleansing achieves the transformation of semantic features from channel to spatial domains, utilizing the correlation of channel information to adequately represent spatial features.

Many existing research works have explored this approach and proposed a series of methods such as ClusDet [35], DMNet [42], GLSAN [43], DMGODet [42], FocusDet [44], and CRENet [45]. Their core components typically include a global object detector, a local object detector, and a foreground region extraction method or network, leading to significant improvements in detecting small targets. However, these methods suffer from one or more drawbacks: they often require introducing additional network structures, increasing the overall model complexity and training difficulty, many involve setting hyperparameters, and the quality of extraction results is greatly influenced by these hyperparameters, resulting in relatively weak overall method stability. Finally, they all adopt the approach of individually sending foreground region blocks to local object detectors for detection, which incurs significant time overhead [46]. Drones use UAV for tasks, such as object detection from the drone's perspective, efficiency is crucial, as detection speed directly impacts the effectiveness of various tasks relying on detection technology [43].

To address these issues with existing global–local fusion detection frameworks, this article introduces a foreground region assembly method. Each extracted foreground region block is scaled according to certain rules and then assembled into a single image, which is input into the local object detector. Finally, the global detection results and local object results are fused using NMS methods to output the final prediction results. The overall framework, as shown in Fig. 5, comprises two object detectors: global and local object detectors. Both detectors use the same detection model proposed in the previous section, which incorporates the expanded detection layer of the YOLOX model to achieve better small object detection performance. The foreground region extraction adopts the

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clustering method proposed by Yu et al. [54] based on the global prediction results. Next, we will detail the methods related to foreground region extraction and assembly, followed by an explanation of the training and inference process of the global–local fusion detection framework based on foreground region assembly.

1) Feature Fusion Guided by Semantic Information: The existing foreground region extraction methods can be broadly categorized into two types: additional foreground region extraction networks and clustering algorithms such as K-means and mean-shift [47]. The former requires adding an extra region extraction network to the overall framework, which increases the complexity of the model structure. In addition, it necessitates manually generating real labels for foreground regions for supervised training, making the entire process cumbersome. On the other hand, the latter clustering algorithms also have their limitations. Some clustering algorithms extract a fixed number of foreground regions, which does not align with the variable number of clustered regions in images captured from the flexible and dynamic perspective of UAV. Moreover, the quality of region blocks obtained through clustering algorithms is unstable—they may easily include too many objects, resulting in large overall block sizes and making it difficult to effectively enhance the detection of small targets. After comprehensive analysis and comparison, this article chooses to adopt the foreground region extraction method proposed by Li et al. [48]. This method adaptively extracts foreground regions of varying quantities based on the predictions of the global detector, as outlined in Algorithm 1.

Algorithm 1: Foreground Region Extraction Algorithm

Input: The globally predicted detection boxes B_g

Output: The collection of foreground region blocks B_r

```
INITIALIZE B_r = ∅
WHILE B_g ≠ ∅ do
    A = argmin(|B'|), B' ∈ B_g
    B_g = B_g − A
    FOR ALL B ∈ B_g do
        IF |A| + |B| > |U|
            A = U
            B_g = B_g − {B}
        END IF
    END FOR
    B_r = B_r ∪ {A}
ENDWhile
```

First, the output of the global detector is subjected to NMS to generate a set of global prediction results and extract a series of corresponding detection boxes. Considering the potential biases in the predictions of the global object detector and to mitigate severe overlap in the final foreground regions, these detection boxes are appropriately enlarged based on their aspect ratios to form the input set B_g.

In the foreground region extraction process, specifically, the following steps are taken: initially, the detection box A with the smallest area in B_g, which has not been clustered, is selected. Then, each remaining detection box $B \in B_g$ is traversed sequentially. For each B, a rectangular region block U that can simultaneously cover both detection boxes A and B is identified. If the sum of the areas of detection boxes A and B is greater than the area of region block U, the detection boxes A and B are merged, with region block U replacing detection box A, and detection box B is removed from B_g. After traversing all detection boxes, A is added to the final set of foreground region blocks B_r. This process is repeated until all initial detection boxes have been clustered.

The aforementioned method efficiently extracts not only high-quality foreground region blocks but it also faces certain challenges. First, it involves setting hyperparameters. In the global detection results, some low-confidence overlapping detection boxes are excluded via NMS. The careful selection of the confidence threshold and IoU threshold in NMS is crucial. If the confidence threshold is set too low, too many detection boxes are inputted into the foreground region extraction method, resulting in excessively large foreground region block sizes containing too many objects, leading to poor block quality and limited enhancement detection effects for small targets. Conversely, setting a high confidence threshold may lead to the omission of some foreground region blocks. Furthermore, subsequent scaling adjustments of foreground region blocks may compress the sizes of initially easily detectable larger targets, increasing the risk of missing or misclassifying them.

Incorporating this clustering method into the global–local fusion detection framework effectively addresses these issues and fully utilizes the advantages of the clustering algorithm. By setting a relatively high confidence threshold, the initial number of global detection boxes is controlled, ensuring that the extracted foreground region blocks have high quality and contain a moderate number of objects. This facilitates the local object detector in achieving good and stable detection results. In addition, the global detector's detection results complement the potential missed true object detection boxes by the local detector and compensate for false detections, thereby improving the overall detection performance and achieving the better enhancement detection for small targets.

2) Foreground Region Matching: Through the foreground region extraction method introduced in the previous section, we obtained a series of foreground region blocks [49]. Many existing global–local fusion detection methods choose to directly adjust the scales of these foreground region blocks and then input them separately into local object detectors [50]. However, the obvious drawback of this strategy is the high computational cost of local detection, as it essentially involves reasoning on multiple image blocks. To save inference time, this section assembles these region blocks into a single image before inputting them into local object detectors for the detection of local foreground region blocks. The overall foreground region assembly process is illustrated in Fig. 6 and mainly consists of two steps: scale adjustment and block assembly. In the scale adjustment step, region blocks containing different-sized targets are relatively scaled according to certain rules to balance the scales of each block, ensuring that small to medium-sized target region blocks, which are not easy to detect, have a relatively larger pixel proportion. In the block assembly step, the prioritized heuristic-based rectangular block assembly method proposed by Zhang et al. [27] is adopted, which efficiently arranges image blocks to minimize the gen-

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eration of unused areas, thereby ensuring that the effective proportion of foreground region blocks in the assembled image is sufficiently large. For scale adjustment, first, calculate the average size s of each foreground region block, which contains the targets

$$s = \frac{1}{N_{\text{Obj}}} \sum_{i=1}^{N_{\text{Obj}}} |\text{obj}_i|$$

where $|\text{obj}_i|$ represents the size of target obj_i and N_{Obj} represents the number of targets contained in the foreground region block. Then, according to the scale division criteria of MS COCO [51], determine the primary size category of the targets contained in the foreground region block for different scale adjustments r. The specific settings are as follows:

$$r = \begin{cases} 4, & \text{if } s < 32 \times 32 \\ 2, & \text{if } 32 \times 32 < s < 96 \times 96 \\ 1, & \text{if } s > 96 \times 96. \end{cases}$$

3) Foreground Region Assembly Method: The previous two sections provided a detailed description of the foreground region extraction and assembly methods relevant to the local detection in the global–local fusion detection framework. This section explains how to utilize foreground region assembly technology for network training and inference.

Regarding the training part, we first train the global object detector using the original training dataset D_t to ensure that the global object detection model exhibits satisfactory detection performance. Then, based on the output results of the global object detector, we generate foreground region assembly images I_t' for each original image I_t using the aforementioned method. These assembled images are organized into a training dataset D_t' for training the local object detector. Ultimately, we obtain a local detection model tailored for foreground region detection. The overall process is illustrated in Fig. 7.

As for the inference part, the main process is outlined in Fig. 5. First, the original aerial images captured by the UAV are input into the global object detector to obtain prediction results R. Then, high-confidence detection boxes are extracted, and overlapping detection boxes are eliminated using nonmaximum the suppression (NMS) method. Subsequently, the corresponding foreground region assembly images I' are generated using the aforementioned method and input into the local object detector to obtain prediction results R' based on I'. These detection boxes are then mapped back to the original images to generate prediction results R'' based on the original images I. Finally, the global detection results R and the locally detected results based on foreground region assembly R'' are fused using the NMS method to obtain the final prediction results.

Mapping the detection boxes based on the assembled foreground region images back to the global image is a crucial step in the inference process, as shown in Algorithm 2. The inputs include the quadruplets of detection boxes based on the assembled foreground images $B_l^j(x, y, w, h)$, where each quadruplet corresponds to the top-left coordinates and the width and height of the detection box, the set of quadruplets for all foreground region blocks $B_r = \{B_r^i(x, y, w, h)\}$, along with their corresponding triplets in the global image set $S_g = \{S_g^i(x, y, r)\}$, where each triplet represents the top-left coordinates and the scaling factor of the region block in the global image, the threshold σ is used to determine which region block the detection box belongs to, set to 0.9 in this article. The output is the detection box of the local detection box mapped back to the global image $B_g^j(x, y, w, h)$. The superscripts j and i in the above denote the indices in the respective sets. Iterate through all the foreground region blocks, calculate the IoU to determine which foreground region block the local detection box belongs to, and then perform the corresponding coordinate transformation and scale adjustment of the detection box size.

Algorithm 2: Detection Box Mapping Algorithm

Input: $B_l(x, y, w, h)$, $U = \{U_i(x, y, w, h)\}$ $S = \{S(x, y, r)\}$, σ

Output: $B_g(x, y, w, h)$

```
FOR ALL U_t ∈ U DO
    IF IoU(U_t, B_l) > σ THEN
        x' = (x_{B_l} − x_{U_t}) * r + x_{S_t}
        y' = (y_{B_l} − y_{U_t}) * r + y_{S_t}
        w' = w_{B_l} * r
        h' = h_{B_l} * r
        B_g(x, y, w, h) = (x', y', w', h')
        BREAK
    END IF
END FOR
```

IV. Experiments and Discussion

This section comprehensively evaluates our self-designed method for integrating semantic guidance features and the global–local fusion detection framework for foreground region assembly. First, the dataset used for experiments is introduced, followed by the evaluation metrics and implementation details. Next, comparative and ablation tests are conducted to qualitatively and quantitatively demonstrate the effectiveness of our designed modules.

A. Experimental Setup

The experiments in this article were conducted using the NVIDIA RTX 3090 GPU with the PyTorch framework. The YOLOX-M model, based on the Modified CSP v5-M as the feature extraction backbone, was used as the base model, with the introduction of a feature fusion guided by semantic information serving as both the global and local object detectors. Extensive and comprehensive experiments were carried out on two benchmark datasets for UAV aerial imagery, namely, VisDrone [52] and UAVDT [53], to validate the effectiveness of the proposed method. Following the validation protocols of previous works, the performance analysis was conducted on the validation dataset of VisDrone [52], and evaluation was performed on the testing dataset of UAVDT [53]. To better assess the detection performance for small and other scale objects, additional evaluation metrics AP_S, AP_M, and AP_L were introduced alongside the standard AP, AP_{50}, and AP_{75} metrics in the previous section. These new metrics correspond to the average precision (AP)

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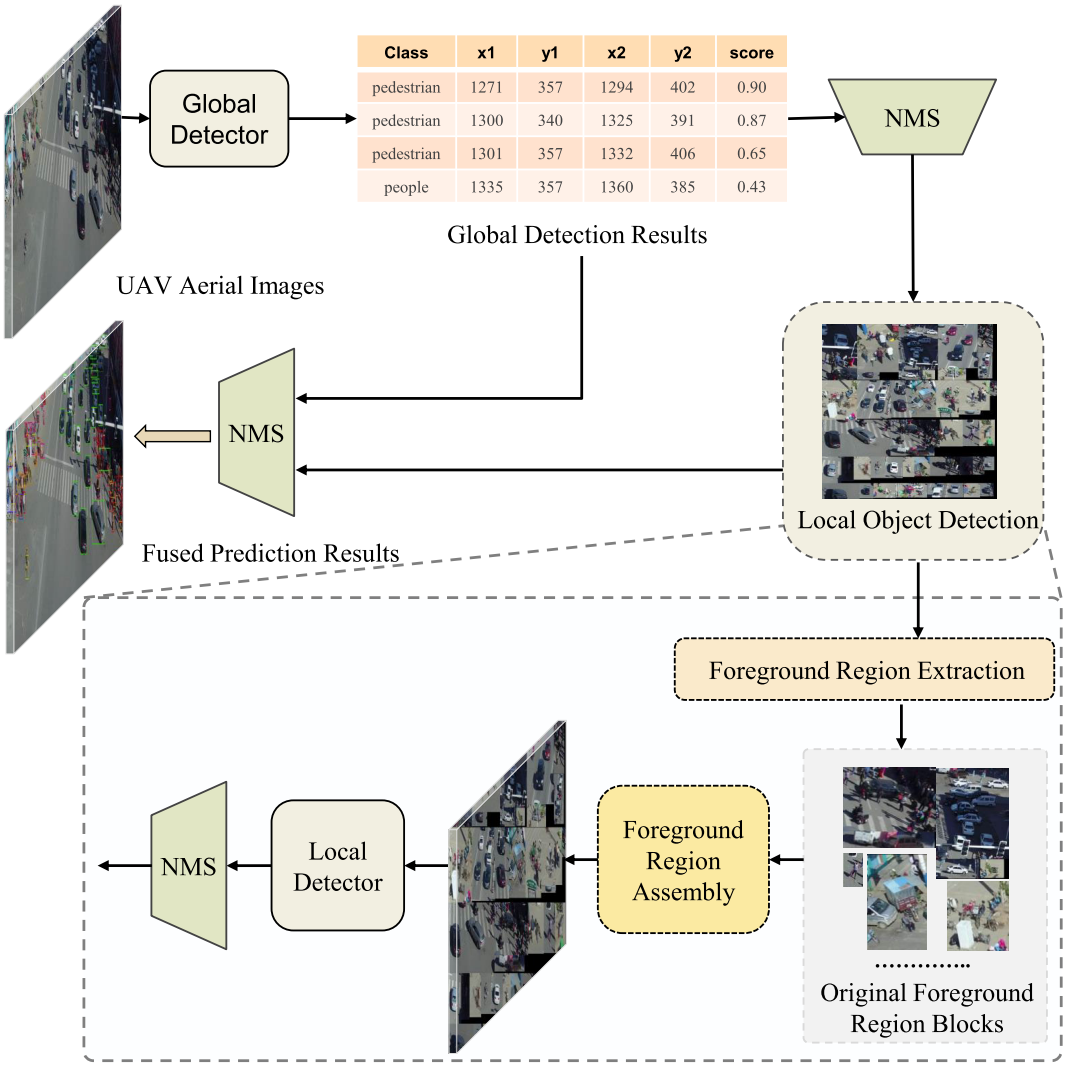


Fig. 5. Aerial images captured by UAV are first processed through a global object detector. The results from the global detector undergo NMS and are then inputted into the region foreground extraction module to obtain region foreground detection blocks. Subsequently, these detection blocks are fed into the foreground region assembly module for assembling foreground regions. Following this, the output is further processed by a local object detector. The results from both the global and local detectors are integrated through comprehensive NMS to yield the final output.

for small, medium, and large objects, respectively, defined according to the MS COCO [51] dataset criteria. During the training phase, pretrained weights based on MS COCO [51] were utilized, along with the Adam optimization strategy. The weight decay was set to 0.0005, and the initial learning rate was set to 0.0001. A cyclic StepLR learning rate strategy was employed. For the VisDrone [52] dataset, the global object detector was trained for 80 epochs, while the local object detector was trained for 40 epochs. For the UAVDT dataset [53], the global object detector was trained for 30 epochs, and similarly, the local object detector was trained for 40 epochs. In addition, the input image size for both the global and local detectors on the two datasets was set to 640× 640 pixels, with each batch containing four images. To more accurately and fairly evaluate the effectiveness of the methods in this section, the Mosaic and Mixup data augmentation strategies inherent in YOLOX were removed during training, retaining only basic image flipping and grayscale adjustments. Regarding model inference, the image size was also set to 640×640 pixels, and only one image was processed for object detection each time.

B. Ablation Experiments

In the case of numerous small targets in UAV images, this article proposes a global–local fusion detection framework based on the foreground region assembly. In addition, a feature fusion guided by semantic information targeting small targets is introduced into the detector. By generating high-resolution, information-rich feature maps, the representation of small targets is enhanced to achieve strengthened detection of small targets and overall improvement in object detection performance from the UAV perspective. In the following sections, we will discuss and analyze the experimental results of the feature fusion guided by semantic information, foreground region assembly-based local object detection, and the final fusion detection to fully demonstrate the effectiveness of these methods.

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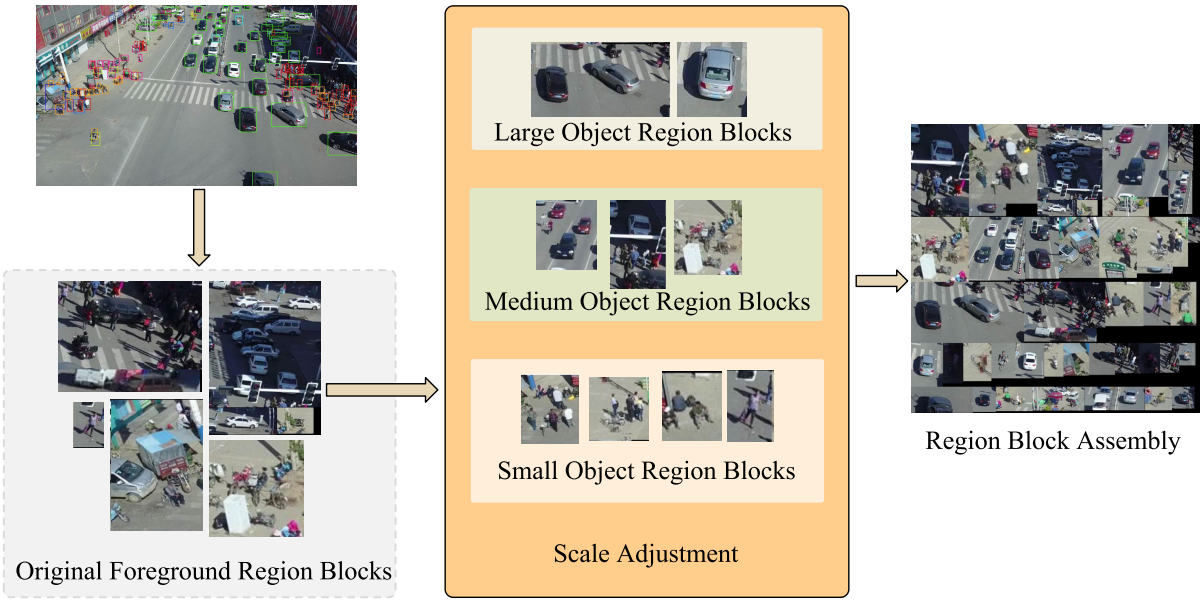


Fig. 6. Output results of the global detector are subjected to NMS processing. Then, the original image regions are divided into regions for detecting large, medium, and small targets. Finally, after scale adjustment, these images are stitched together to form assembled foreground images.

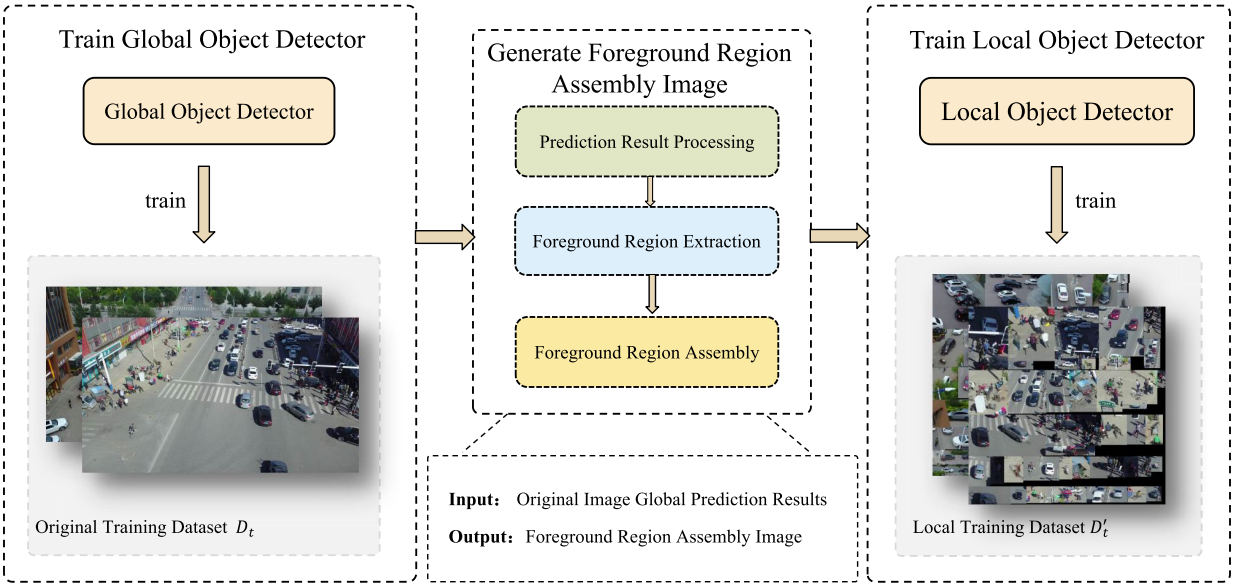


Fig. 7. This classic residual structure takes input X, passes it through average pooling, followed by fully connected layers, an activation function, another fully connected layer, and a sigmoid function to obtain probabilities, which are then multiplied with X. Finally, the result is inputted again for residual stacking to produce the output.

First, this article verifies the effectiveness of the feature fusion guided by semantic information on the VisDrone [52] and UAVDT datasets [53], as shown in Table I, where the values are presented in percentage. The addition of the feature fusion guided by semantic information to the base model is indicated by "+SFM." It can be observed that after introducing the feature fusion guided by semantic information, all performance metrics show improvement. Specifically, on the VisDrone dataset [52], the AP and AP_{50} increased by 0.6% and 1.5%, respectively, while on the UAVDT dataset [53], they increase by 1.2% and 1.2%. Moreover, the AP for small targets (AP_S) increases by 0.6% and 0.5% on the VisDrone [52] and UAVDT datasets [53], respectively. This demonstrates that integrating the feature fusion guided by semantic information into the model enhances the detection performance for small targets and improves the overall detection accuracy.

To better detect small targets in UAV images, the foreground region assembly technology is introduced to detect local foreground region blocks. By extracting foreground region blocks and adjusting their scales, the blocks containing medium and small-sized targets are relatively enlarged, reducing the proportion of small targets and thereby lowering the difficulty of detecting these targets with fewer pixel coverage in the original image. Table II lists the distribution proportions of various scale category targets before and after introducing foreground

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TABLE I
ABLATION EXPERIMENTS OF THE FEATURE FUSION MODULE GUIDED BY SEMANTIC INFORMATION (%)

Dataset	Method	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
VisDrone	YOLOX-M	27.6	47.7	27.5	17.6	41.0	46.0
	+SFM	28.2	49.2	27.8	18.2	41.2	46.4
UAVDT	YOLOX-M	15.8	27.7	16.3	10.0	27.3	24.8
	+SFM	17.0	28.9	18.1	10.5	29.0	24.8

TABLE II
COMPARISON OF SCALE CATEGORY TARGET PROPORTIONS BEFORE AND AFTER INTRODUCING FOREGROUND REGION ASSEMBLY TECHNOLOGY (%)

Dataset	Method	Small Objects	Medium Objects	Large Objects
VisDrone-train	-	60.49	33.98	5.53
	FRP	3.29	58.75	37.96
VisDrone-val	-	68.6	28.65	2.75
	FRP	4.13	66.16	29.71
UAVDT-train	-	50.27	48.34	1.39
	FRP	0.8	70.13	29.07
UAVDT-test	-	74.9	22.99	2.11
	FRP	0.99	67.13	31.88

region assembly technology on the VisDrone [52] and UAVDT datasets [53], providing a more intuitive understanding of this change. The term foreground region packing (FRP) in the table denotes the use of the foreground region assembly method. The primary purpose of adopting foreground assembly technology is to transform the original UAV images from a distribution primarily consisting of small targets to one with medium to large-sized targets, which significantly reduces the task difficulty of local target detectors and thereby achieves better detection performance on the original images.

To thoroughly verify the effectiveness of the local detection method based on foreground region assembly in improving small object detection, this article conducted ablation experiments on the VisDrone dataset [52]. Based on the predictions of the global object detector, foreground region assembly images were generated and local object detection was performed. The evaluation results below are obtained by mapping the detection results on the foreground assembly images back to the original images, ensuring the fairness and reliability of the entire evaluation process. For better comparison, both global and local object detectors utilized the same network structure. This article selected the base model YOLOX-M and the model with the introduced feature fusion guided by semantic information and conducted relevant comparative experiments on both detectors to fully demonstrate that the foreground region assembly technology significantly reduces the proportion of small targets, thereby reducing the detection difficulty of these targets for the object detector and improving overall performance. The final experimental results are shown in Table III, where all comprehensive indicators have significantly improved, especially AP_S has been

TABLE III
ABLATION EXPERIMENT OF FUSION DETECTION BASED ON FOREGROUND REGION ASSEMBLY TECHNOLOGY (%)

Object Detector	Method	AP	Ap_{50}	Ap_{70}	Ap_S	Ap_M	Ap_L
YOLOX-M	-	27.6	47.7	27.5	17.6	41.0	46.0
	FRP	29.6	50.3	30.4	21.2	41.0	46.3
	Fusion	31.0	52.6	31.7	22.0	42.8	47.4
+SFMr	-	28.2	49.2	27.8	18.2	41.2	46.4
	FRP	30.7	52.2	31.7	22.3	41.4	48.5
	Fusion	32.0	54.2	32.8	23.3	42.9	50.1

greatly enhanced. Using YOLOX-M as the object detector, AP, AP_{50}, and AP_{75} have increased by 2.0%, 2.6%, and 2.9%, respectively, and the indicator AP_S for small object detection has increased by 3.6%. In addition, using YOLOX-M with the introduced feature fusion guided by semantic information as the object detector, the three indicators AP, AP_{50}, and AP_{75} have increased by 2.5%, 3.0%, and 3.9%, respectively, and the average detection accuracy for small targets has reached 22.3%. These results strongly validate that the local object detection method based on foreground region assembly effectively mitigates the problem of insufficient feature representation and difficulty in detecting small targets. Furthermore, this method significantly enhances the overall detection performance of small-target-dominated aerial images from the perspective of UAV. Moreover, it is noted that introducing the feature fusion guided by semantic information in the object detector helps improve the quality of detection, thereby further confirming the rationality and effectiveness of the proposed feature fusion guided by semantic information. This article adopts the global–local fusion detection framework to obtain the final prediction results, where the detection boxes from the local object detector are mapped back to the original image, and the global detection results are fused using NMS method. If local detection is solely based on foreground assembly images, inevitably, some misses and false alarms may occur. On one hand, since the extraction of foreground region blocks is based on the predictions of the global object detector, a confidence threshold is set during this process to ensure a moderate number of objects in each foreground region block. This may result in the removal of some low-score global detection boxes, which could potentially contain true objects. However, these corresponding objects do not exist in the local images and therefore cannot be detected, leading to false negatives. On the other hand, to enhance the detection of originally small-sized targets by the local detector, scale adjustment is performed, causing the originally large targets to relatively shrink, making it difficult to effectively identify some larger targets in the original image.

By observing the data in the third and sixth rows of Table III, it can be clearly seen that adopting the global–local fusion detection strategy can effectively improve the overall detection quality. There is a significant improvement in detecting small targets, and there is also an increase in the detection accuracy of targets of other sizes. Specifically, when using YOLOX-M as the object detector, compared with only conducting local detection, the AP, AP_{50}, and AP_{75} are increased by 1.4%, 2.3%, and 1.3%, respectively, with a 0.8% improvement in the small target metric. When using YOLOX-M with

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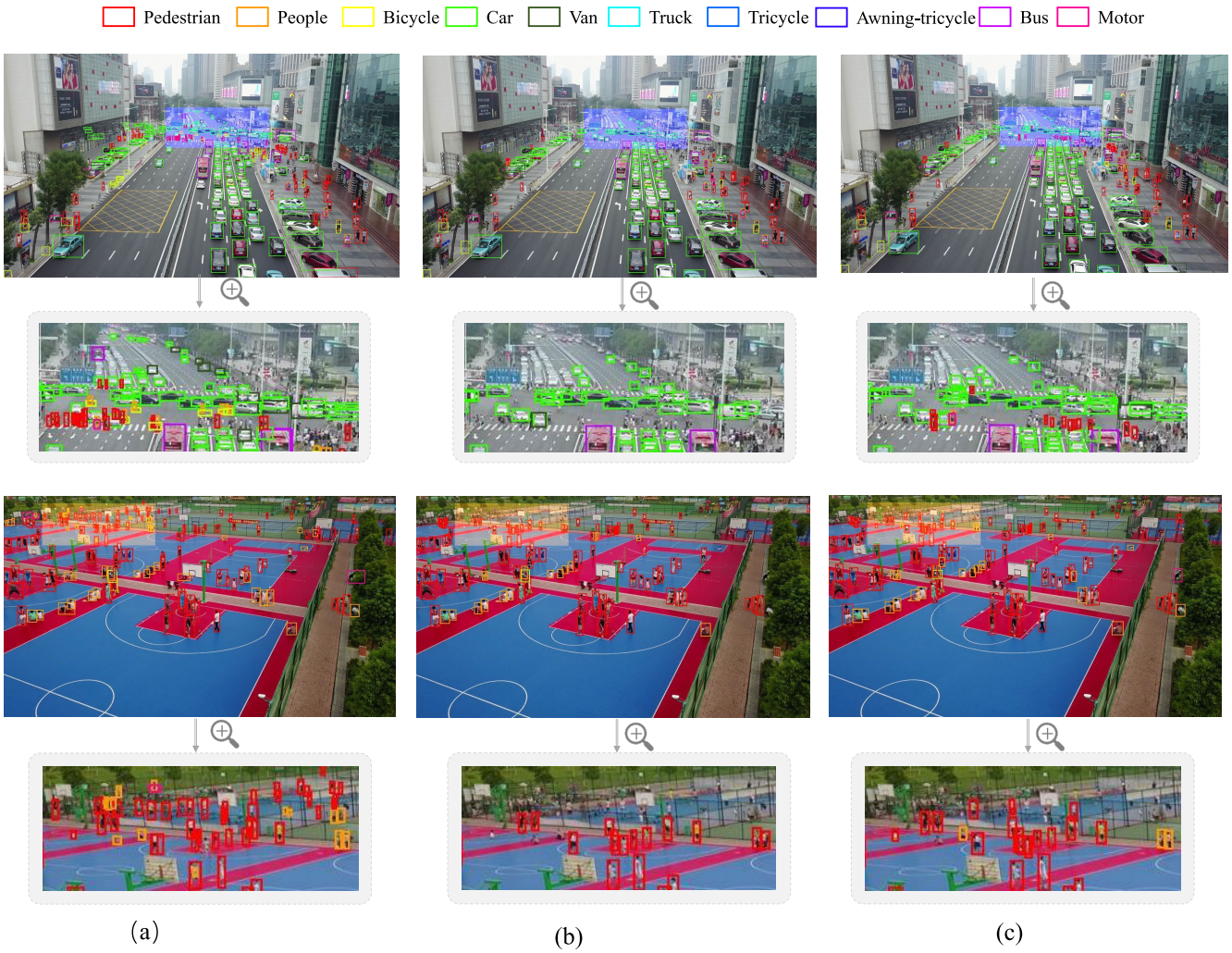


Fig. 8. (a) Annotation of the dataset. (b) Detection performance of YOLOX-M. (c) Detection performance of our model. By comparing with YOLOX-M, we can clearly see that the improvement of our model in detecting small objects is significant, especially in the detection of small crowd targets, where our recall rate is much higher than that of YOLOX-M.

TABLE IV
ABLATION EXPERIMENTS (%) ON THE DETECTOR ENSEMBLE SELECTION IN FUSION DETECTION (%)

Det_g	Det_l	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
YOLOX-M	YOLOX-M	31.0	52.6	31.7	22.0	42.8	47.4
+SFM	YOLOX-M	31.6	53.7	32.0	22.9	42.8	50.3
+SFM	+SFM	32.0	54.2	32.8	23.3	42.9	50.1

the introduced feature fusion guided by semantic information as the object detector, the metrics are increased by 1.3%, 2.0%, and 1.1%, respectively. These experimental results strongly indicate the necessity of global–local fusion detection and validate the analysis in this article regarding the shortcomings of conducting only local detection.

Finally, in pursuit of better detection performance, this section also conducted ablation experiments on the selection of global and local object detectors to find the optimal combination, as shown in Table IV, where Det_g represents the global object detector, and Det_l corresponds to the local object detector. The results indicate that introducing the feature fusion guided by semantic information to both global and local object detectors yields the best detection accuracy. The comprehensive metrics AP, AP_{50}, and AP_{75} reach 32.0%, 54.2%, and 32.8%, respectively, with the most crucial small object detection metric achieving 23.3%, which is a 5.7% improvement compared with the initial YOLOX-M detector. The enhancement is significant, providing strong evidence for the important role of the feature fusion guided by semantic information in strengthening small object detection. Through the extensive quantitative ablation experiments described above, this article thoroughly demonstrates the necessity and effectiveness of the proposed feature fusion guided by semantic information, the local detection based on foreground region assembly, and the global–local fusion detection scheme. To better illustrate the enhancement of small object detection in UAV aerial images achieved by the proposed solution, this section also selects two representative and challenging images from the VisDrone dataset [52]. By visualizing the actual detection results of the baseline model and the proposed method, as shown in Fig. 8(b) and (c), the effectiveness of the approach is vividly demonstrated. In Fig. 8, we present an image captured by a UAV of a main urban road, featuring bustling traffic and a wide variety of complex objects, making the scene rich and diverse. In addition, the image was

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TABLE V
COMPARISON RESULTS ON THE VISDRONE AND UAVDT DATASETS, AGAINST GENERAL DETECTION MODELS AS WELL AS THE CURRENT OPTIMAL GLOBAL AND LOCAL MODELS

Dataset	Method	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	Params(M)	FPS(M)	FLOPs(A)
VisDrone	YOLOX-M	27.6	47.7	27.5	17.6	41	46.1	25.3	67.2	73.8
	YOLOv8	28.5	49.2	28.3	19.5	40.3	46.5	25.9	72.8	78.9
	YOLOv9	29.5	49.9	29.4	20.2	41.7	47.8	20.1	67.4	76.8
	YOLOv10	29.1	49.6	29	19.9	41.2	47.2	15.4	69.2	59.1
	YOLOv11	30.2	50.7	30.1	20.7	42.8	49.1	20.1	74.6	68.1
	ClusDet	30.5	51.3	30.2	20.9	43.2	49.4	-	41.5	-
	DMNet	31.1	52.7	31	21.2	42.9	50.1	-	45.2	-
	Ours	32.3	54.2	32.8	23.3	42.9	50.2	60.9	48.6	100.4
UAVDT	YOLOX-M	15.8	27.7	16.3	10	27.3	24.8	25.3	71.6	73.8
	YOLOv8	16.2	28.2	16.8	10.7	28.1	25.3	25.9	73.2	78.9
	YOLOv9	16.1	28.1	16.5	10.6	27.9	25.3	20.1	70.6	76.8
	ClusDet	16.5	28.6	16.9	11.1	28.6	25.9	-	50.6	-
	YOLOv10	16.6	28.8	17.1	11.2	28.7	26.3	15.4	74.5	59.1
	DMNet	17.1	29.1	17.2	11.5	28.9	26.3	-	52.4	-
	YOLOv11	17.6	29.3	17.3	11.6	29.1	26.4	20.1	76.8	68.1
	Ours	18.3	29.8	17.6	11.8	29.3	26.8	60.9	56.7	100.4

captured with a wide field of view, revealing densely packed small objects in the distance. To better illustrate, this article highlights areas with dense clusters of small objects in a blue transparent box and magnifies this area for better observation. It can be observed that the baseline model YOLOX-M has poor detection capabilities for distant small objects, exhibiting numerous missed detections, particularly struggling to detect pedestrians with minimal pixel coverage and occasionally misclassifying vehicle categories. In contrast, our proposed method demonstrates improved overall detection performance, effectively recognizing a significant portion of the dense crowd walking on the pedestrian crossing.

C. Comparison With State-of-the-Art Methods

In the comparative experiments, we conducted a comprehensive analysis of our proposed method against the current state-of-the-art algorithms on two public datasets, VisDrone and UAVDT. The VisDrone dataset includes various UAV-captured scenarios in complex urban environments, while the UAVDT dataset primarily focuses on traffic and crowd scenes, each presenting unique challenges.

On the VisDrone dataset, our method demonstrated excellent performance across multiple key metrics compared with the current best methods. Specifically, our model achieved a 4.1% improvement in AP, and improvements of 3.8% and 3.5% in AP_{50} and AP_{75}, respectively. For target detection at different scales, our method improved by 4.5%, 3.9%, and 3.2% for small AP_S, medium AP_M, and large targets AP_L, respectively. These results indicate that our model performs well across different scales, complex occlusions, and dense scenes, effectively enhancing overall target detection performance.

On the UAVDT dataset, our method also showed significant superiority over the current best methods. Our method achieved a 3.7% improvement in AP, and improvements of 3.4% and 3.1% in AP_{50} and AP_{75}, respectively. For small AP_S, medium AP_M, and large targets AP_L, our method improved by 4.0%, 3.5%, and 3.0% across these metrics. These improvements are attributed to our model's structural design, which enhances small target features and leverages contextual information, resulting in more robust performance, particularly in detecting fast-moving and occluded targets in complex traffic scenes.

In summary, our method demonstrates significant advantages over the current state-of-the-art methods across multiple metrics, including AP, AP_{50}, AP_{75}, AP_S, AP_M, and AP_L on both the VisDrone and UAVDT datasets. Notably, our method outperforms existing state-of-the-art methods in challenging scenarios involving small target detection, high occlusion, and complex backgrounds, highlighting the effectiveness and robustness of our model. For detailed comparative experimental results, please refer to Table V.

V. CONCLUSION AND DISCUSSION

UAV aerial images typically contain numerous small objects. Due to the limited number of pixels occupied by small objects and their susceptibility to noise interference, their feature representations may be insufficient during the process of feature extraction and fusion. Compared with larger scale objects, detecting small objects is more challenging, posing significant obstacles to target detection from the perspective of UAV imagery. The model still has issues with detecting distant small objects. In the future, we will explore infrared small object detection methods to address this problem. This article primarily focuses on enhancing the detection of small objects in UAV aerial images. It analyzes and summarizes existing research approaches and methods, initiates discussions, conducts research, and proposes a global–local fusion detection framework based on foreground region assembly technology. In addition, a feature fusion module guided by semantic information is introduced into the detector to achieve better detection results for small objects. The incorporation of extended detection layers for small objects and the guidance of subpixel semantic information have increased model com-

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plexity and slowed inference speed. Future work will focus on exploring lightweight model designs.

First, this article introduces the feature fusion module guided by semantic information. The core idea is to effectively utilize contextual information of different scales to generate high-resolution feature maps containing sufficient semantic and detailed information without changing the input size of the image, thereby enhancing the feature representation of small objects at that detection layer. Simultaneously, this layer has a relatively smaller receptive field, which facilitates more precise target localization and extraction of small object texture information. Moreover, in the UAV domain, many researchers tackle the problem of small object detection using a global–local fusion detection approach. However, their solutions suffer from issues, such as complex network structures, difficulty in training, unstable results, and high inference time costs. Therefore, this article proposes a global–local fusion detection framework based on foreground region assembly technology, assembling foreground region blocks into a single image. The entire inference process only requires two inferences—global and local, significantly improving detection efficiency. Furthermore, appropriate clustering methods are employed to ensure the quality of the extracted foreground region blocks, thereby enhancing the detection performance of local targets. Finally, through quantitative multiple ablation experiments and visualization results, the rationality and effectiveness of the proposed method are convincingly demonstrated. In addition, significant accuracy improvements are achieved on two UAV aerial image target detection benchmark datasets, VisDrone [52] and UAVDT [53], compared with other similar algorithms, with higher efficiency.

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Yaxiong Chen (Member, IEEE) received the B.Sc. degree in mathematics from Hubei University, Wuhan, China, in 2014, the M.Sc. degree in mathematics from Wuhan University of Technology, Wuhan, in 2017, and the Ph.D. degree in signal and information processing from the University of Chinese Academy of Sciences, Beijing, China, in 2020.

He is currently a Professor with the School of Computer Science and Artificial Intelligence, Wuhan University of Technology. His research interests include pattern recognition, machine learning, hyperspectral image analysis, and medical imaging.



Zhengze Ye received the B.Eng. degree from Qingdao University of Technology, Qingdao, China, in 2022. He is currently pursuing the master's degree with the School of Computer and Artificial Intelligence, Wuhan University of Technology, Wuhan, China.

His research interests include low-altitude UAV target detection and robust adversarial detection in remote sensing imagery.



Haokai Sun received the B.S. degree in computer science and technology from Wuhan University of Technology, Wuhan, China, in 2021, where he is currently pursuing the Master of Engineering degree in computer science and technology with the School of Computer Science and Artificial Intelligence.

His research interests include machine learning and pattern recognition.



Tengfei Gong received the Ph.D. degree from the Key Laboratory of Spectral Imaging Technology, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, China, and the University of Chinese Academy of Sciences, Beijing, China, in June 2022.

She is currently an Associate Professor with the School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan, China. Her research interests include pattern recognition, computer vision, and machine learning.



Shengwu Xiong received the B.Sc. degree in computational mathematics and the M.Sc. and Ph.D. degrees in computer software and theory from Wuhan University, Wuhan, China, in 1987, 1997, and 2003, respectively.

He is currently a Professor with the School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan. His research interests include intelligent computing, machine learning, and pattern recognition.



Xiaoqiang Lu (Senior Member, IEEE) is currently a Full Professor with the College of Physics and Information Engineering, Fuzhou University, Fuzhou, China.

His research interests include intelligent optical sensing, pattern recognition, machine learning, and hyperspectral image analysis.

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